A2. Data preparation – selection and cleaning

Data selection refers to choose the most relevant attributes or samples. Meanwhile, data cleaning can be accomplished by applying filters to the data in the Preprocess tab.

1. Start with the Preprocess tab. Study numeric attributes. Give the mean, min & max and its standard deviation.

- Attribute: CustomerID

Statistic	Value
Minimum	50001
Maximum	55630
Mean	52815.5
Standard Deviation	1625.385

- Attribute: Tenure

Statistic	Value
Minimum	0
Maximum	61
Mean	9.759
Standard Deviation	8.577

- Attribute: CityTier

Statistic	Value
Minimum	1
Maximum	3

Mean	1.655
Standard Deviation	0.915

- Attribute: WarehouseToHome

Statistic	Value
Minimum	5
Maximum	127
Mean	15.64
Standard Deviation	8.531

- Attribute: HourSpendOnApp

Statistic	Value
Minimum	0
Maximum	5
Mean	2.799
Standard Deviation	0.932

- Attribute: NumberOfDeviceRegistered

Statistic	Value
Minimum	1
Maximum	6
Mean	3.689
Standard Deviation	1.024

- Attribute: SatisfactionScore

Statistic	Value
Minimum	1
Maximum	5
Mean	3.067
Standard Deviation	1.38

- Attribute: NumberOfAddress

Statistic	Value
Minimum	1
Maximum	22
Mean	4.214
Standard Deviation	2.584

- Attribute: Complain

Statistic	Value
Minimum	0
Maximum	1
Mean	0.285
Standard Deviation	0.451

- Attribute: OrderAmountHikeFromlastYear

Statistic	Value
Minimum	11
Maximum	26
Mean	15.708
Standard Deviation	3.675

- Attribute: CouponUsed

Statistic	Value
Minimum	0
Maximum	16
Mean	1.751
Standard Deviation	1.895

- Attribute: OrderCount

Statistic	Value
Minimum	1
Maximum	16
Mean	2.916
Standard Deviation	2.902

- Attribute: DaySinceLastOrder

Statistic	Value

Minimum	0
Maximum	46
Mean	4.296
Standard Deviation	3.7

- Attribute: CashbackAmount

Statistic	Value
Minimum	0
Maximum	325
Mean	177.221
Standard Deviation	49.194

- Attribute: AvgCashbackPerOrder

Statistic	Value
Minimum	0
Maximum	299
Mean	101.114
Standard Deviation	61.531

- Attribute: EngagementScore

Statistic	Value
Minimum	0
Maximum	320

Mean	26.419
Standard Deviation	35.16

- Activity: InactivityRadio

Statistic	Value
Minimum	0
Maximum	15
Mean	1.011
Standard Deviation	1.586

- Attribute: LoyaltyScore

Statistic	Value
Minimum	0
Maximum	465
Mean	32.666
Standard Deviation	55.22

- Attribute: ChurnCategory

Statistic	Value
Minimum	0
Maximum	2
Mean	0.634
Standard Deviation	0.754

2. Study the nominal attributes and report the values of each attribute and the count of each.
    - Attribute: PreferredLoginDevice

No.	Label	Count
1	Mobile Phone	2765
2	Phone	1231
3	Computer	1634

    - Attribute: PreferredPaymentMode

No.	Label	Count
1	Debit Card	2314
2	UPI	414
3	CC	273
4	Cash on Delivery	149
5	E wallet	614
6	COD	365
7	Credit Card	1501

    - Attribute: Gender

No.	Label	Count
1	Female	2246
2	Male	3384

    - Attribute: PreferredOrderCat

No.	Label	Count
1	Laptop & Accessory	2050
2	Mobile	809
3	Mobile Phone	1271

4	Others	264
5	Fashion	826
6	Grocery	410

- Attribute: MaritalStatus

No.	Label	Count
1	Single	1796
2	Divorced	848
3	Married	2986

3. Identify the attributes with missing/noise values. Remove the missing/noise values with the method of your choice using WEKA, explaining which filter you are using and why you make this choice.

- Attribute: WarehouseToHome

  Total data missing: 251 (4%)

  After ReplaceMissingValues: 0 (0%)

- Attribute: OrderAmountHikeFromlastYear

  Total data missing: 265 (5%)

  After ReplaceMissingValues: 0 (0%)

- Attribute: CouponUsed

  Total data missing: 256 (5%)

  After ReplaceMissingValues: 0 (0%)

- Attribute: OrderHikeRate

  Total data missing: 508 (9%)

  After ReplaceMissingValues: 0 (0%)

- Attribute: InactivityRatio

  Total data missing: 508 (9%)

  After ReplaceMissingValues: 0 (0%)

Filter used: ReplaceMissingValues

The ReplaceMissingValues filter in WEKA was selected because it provides a simple yet effective method for handling missing data without removing entire records. This filter automatically replaces missing values based on the attribute type:

- For numeric attributes, it replaces missing values with the mean of the attribute.
- For nominal (categorical) attributes, it replaces missing values with the mode (most frequent value).

This approach preserves the dataset's overall structure and statistical distribution, which is especially important when dealing with a limited number of instances. By imputing missing values instead of deleting rows or columns, we reduce data loss and maintain the integrity of the dataset for further analysis or modeling.

Additionally, this filter is built into WEKA, easy to apply, and does not require external configuration, making it a practical choice for efficient and consistent data cleaning.

4. Identify the attributes with outliers. Investigate the method for detecting outliers. Are there any outliers in this dataset, and if yes, describe how you deal with these outliers.

    1. Apply Interquartile range filter
    2. identify the outliers and extreme values

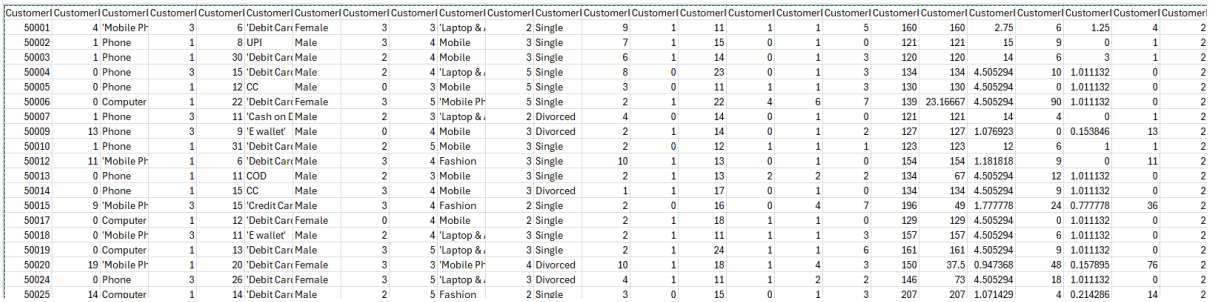
outliers	extreme values
1092	329

    3. handle outliers and extreme by removing instances

        filter > unsupervised > instance > removewithvalues

    4. From 5630 reduce to 4447

5. Save the cleaned dataset into file-cleaned.arff. Show several samples in the dataset, at least the first 20 rows of this dataset – with the columns.



Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer	Customer
50001	4	'Mobile Ph	3	6	'Debit Car	Female	3	3	'Laptop &	2	Single	9	1	11	1	1	5	160	160	2.75	6	1.25	4	2
50002	1	Phone	1	8	UPI	Male	3	4	Mobile	3	Single	7	1	15	0	1	0	121	121	15	9	0	1	2
50003	1	Phone	1	30	'Debit Car	Male	2	4	Mobile	3	Single	6	1	14	0	1	3	120	120	14	6	3	1	2
50004	0	Phone	3	15	'Debit Car	Male	2	4	'Laptop &	5	Single	8	0	23	0	1	3	134	134	4.505294	10	1.011132	0	2
50005	0	Phone	1	12	CC	Male	0	3	Mobile	5	Single	3	0	11	1	1	3	130	130	4.505294	0	1.011132	0	2
50006	0	Computer	1	22	'Debit Car	Female	3	5	'Mobile Ph	5	Single	2	1	22	4	6	7	139	23.16667	4.505294	90	1.011132	0	2
50007	1	Phone	3	11	'Cash on D	Male	2	3	'Laptop &	2	Divorced	4	0	14	0	1	0	121	121	14	4	0	1	2
50009	13	Phone	3	9	'E wallet'	Male	0	4	Mobile	3	Divorced	2	1	14	0	1	2	127	127	1.076923	0	0.153846	13	2
50010	1	Phone	1	31	'Debit Car	Male	2	5	Mobile	3	Single	2	0	12	1	1	1	123	123	12	6	1	1	2
50012	11	'Mobile Ph	1	6	'Debit Car	Male	3	4	Fashion	3	Single	10	1	13	0	1	0	154	154	1.181818	9	0	11	2
50013	0	Phone	1	11	COD	Male	2	3	Mobile	3	Single	2	1	13	2	2	2	134	67	4.505294	12	1.011132	0	2
50014	0	Phone	1	15	CC	Male	3	4	Mobile	3	Divorced	1	1	17	0	1	0	134	134	4.505294	9	1.011132	0	2
50015	9	'Mobile Ph	3	15	'Credit Car	Male	3	4	Fashion	2	Single	2	0	16	0	4	7	196	49	1.777778	24	0.777778	36	2
50017	0	Computer	1	12	'Debit Car	Female	0	4	Mobile	2	Single	2	1	18	1	1	0	129	129	4.505294	0	1.011132	0	2
50018	0	'Mobile Ph	3	11	'E wallet'	Male	2	4	'Laptop &	3	Single	2	1	11	1	1	3	157	157	4.505294	6	1.011132	0	2
50019	0	Computer	1	13	'Debit Car	Male	3	5	'Laptop &	3	Single	2	1	24	1	1	6	161	161	4.505294	9	1.011132	0	2
50020	19	'Mobile Ph	1	20	'Debit Car	Female	3	3	'Mobile Ph	4	Divorced	10	1	18	1	4	3	150	37.5	0.947368	48	0.157895	76	2
50024	0	Phone	3	26	'Debit Car	Female	3	5	'Laptop &	3	Divorced	4	1	11	1	2	2	146	73	4.505294	18	1.011132	0	2
50025	14	Computer	1	14	'Debit Car	Male	2	5	Fashion	2	Single	3	0	15	0	1	3	207	207	1.071429	4	0.214286	14	2

A3. Data preparation – transformation

Among the different data transformation techniques, explore those available through the WEKA filters in the Preprocess tab. Study the following data transformation and report which you have applied:

1. Perform normalization when necessary. Explain which filter that you applied; Min-max normalization, Z-score normalization or Decimal normalization and provide detailed information the method of your choice – state which one you choose and why. *You may not require to normalize all attributes. Explain why you do the normalization on the attributes.

- **Perform normalization using normalization.**
    - Attribute:Tenure

Statistic	Value

Minimum	0
Maximum	1
Mean	0.203
Standard Deviation	0.165

- Attribute:CityTier

Statistic	Value
Minimum	0
Maximum	1
Mean	0.328
Standard Deviation	0.458

- Attribute: WarehouseToHome

Statistic	Value
Minimum	0
Maximum	1
Mean	0.34
Standard Deviation	0.259

- Attribute: HourSpendOnApp

Statistic	Value
Minimum	0
Maximum	1

Mean	0.683
Standard Deviation	0.24

- Attribute: NumberOfDeviceRegistered

Statistic	Value
Minimum	0
Maximum	1
Mean	0.529
Standard Deviation	0.203

- Attribute: SatisfactionScore

Statistic	Value
Minimum	0
Maximum	1
Mean	0.5
Standard Deviation	0.343

- Attribute: NumberOfAddress

Statistic	Value
Minimum	0
Maximum	1
Mean	0.327
Standard Deviation	0.257

- Attribute: Complain

Statistic	Value
Minimum	0
Maximum	1
Mean	0.28
Standard Deviation	0.449

- Attribute: OrderAmountHikeFromlastYear

Statistic	Value
Minimum	0
Maximum	1
Mean	0.298
Standard Deviation	0.231

- Attribute: CouponUsed

Statistic	Value
Minimum	0
Maximum	1
Mean	0.267
Standard Deviation	0.215

- Attribute: OrderCount

Statistic	Value
Minimum	0

Maximum	1
Mean	0.141
Standard Deviation	0.193

- Attribute: DaySinceLastOrder

Statistic	Value
Minimum	0
Maximum	1
Mean	0.218
Standard Deviation	0.191

- Attribute: CashbackAmount

Statistic	Value
Minimum	0
Maximum	1
Mean	0.539
Standard Deviation	0.145

- Attribute: AvgCashbackPerOrder

Statistic	Value
Minimum	0
Maximum	1
Mean	0.376

Standard Deviation	0.2

- Attribute: OrderHikeRate

Statistic	Value
Minimum	0
Maximum	1
Mean	0.212
Standard Deviation	0.236

- Attribute: EngagementScore

Statistic	Value
Minimum	0
Maximum	1
Mean	0.184
Standard Deviation	0.173

- Attribute: InactivityRatio

Statistic	Value
Minimum	0
Maximum	1
Mean	0.192
Standard Deviation	0.202

- Attribute: LoyaltyScore

Statistic	Value
Minimum	0
Maximum	1
Mean	0.172
Standard Deviation	0.186

- **Attributes that are excluded from normalization**

Attributes	**Reason**
- ChurnCategory	Class Attribute
- PreferredLoginDevice - PreferredPaymentMode - Gender - PreferedOrderCat - MartialStatus	Non-Numeric

2. Perform discretization when necessary. Which attributes and how many bins have you implemented? Explain.

- For this task, I did not discretize the numeric attributes. Nevertheless, carried out data type transformation by changing a numeric attribute “ChurnCategory” to nominal.
- Using filter RanemNominalValues. Change label from 0,1,2 to NotChurn, Uncertain,Churn.
- Attribute: ChurnCategory

No.	Label	Count	Weight
1	NotChurn	2376	2376
2	Uncertain	1432	1432
3	Churn	639	639

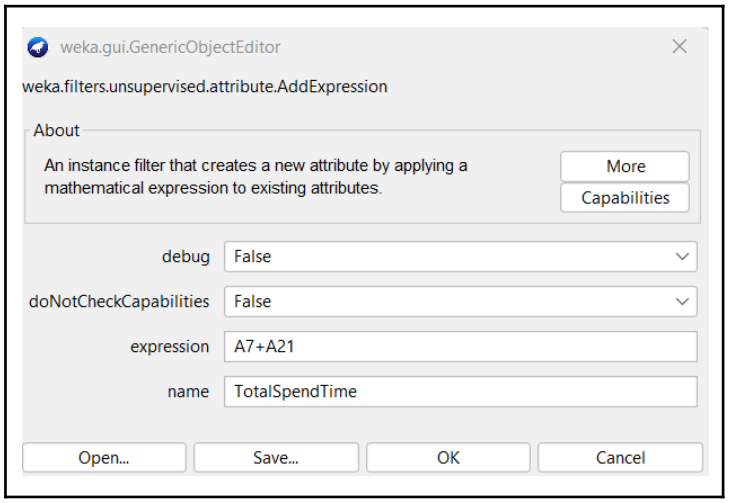
3. Perform attribute construction when necessary – for example adding an attribute representing the sum of two other ones. Which Weka filter permits to do this? Show the attribute if you applied it.

- Create new attribute:
  TotalSpendTime= HourSpendOnApp + EngagementScore

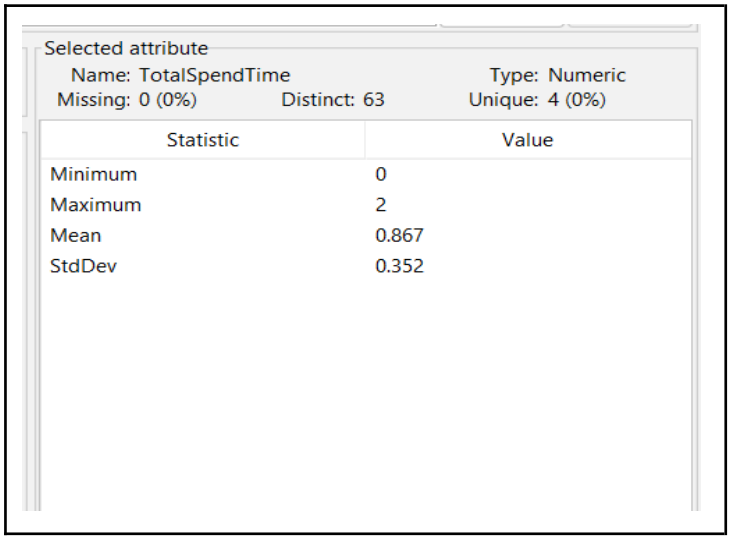
1. Choose filter

   filters > unsupervised > attribute > AddExpression

2. In the AddExpression. Change expression to A7+A21 then name the attribute as TotalSpendTime.



3. New attribute has completed.

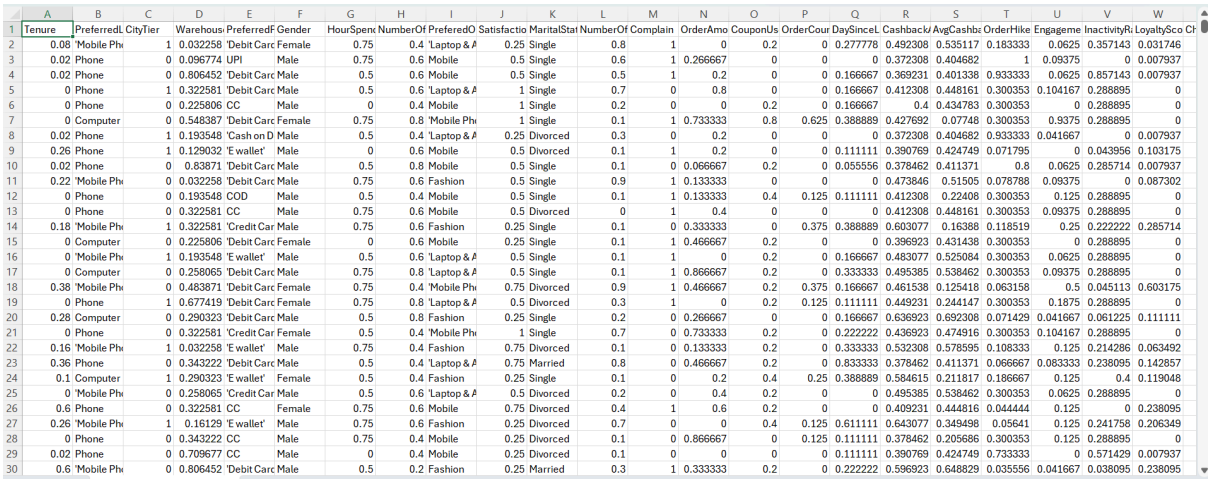


4. Perform other specific processes when necessary and explain.

- There was nothing significant so had to preprocess in doing this task because the dataset was already clean and ready to use.
- Having analyzed the data, managed to ensure that:
  - Attributes that were required to be removed had no identifiers.
  - There was no missing value that was required to be imputed.
  - Every characteristic was pertinent and organized well, with appropriate data types.

- The target attribute ChurnCategory was properly transformed to nominal.

5. Save the prepared dataset into file_processed.arff. Show the samples, at least the first 20 rows of this dataset – with all columns.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
1	Tenure	PreferredL	CityTier	Warehous	PreferredF	Gender	HourSpen	NumberOf	PreferedO	Satisfactio	MaritalStat	NumberOf	Complain	OrderAmo	CouponUs	OrderCour	DaySinceL	Cashback	AvgCashb	OrderHike	Engageme	InactivityRi	LoyaltySco	Cl
2	0.08	'Mobile Ph	1	0.032258	'Debit Carc	Female	0.75	0.4	'Laptop & A	0.25	Single	0.8	1	0	0.2	0	0.277778	0.492308	0.535117	0.183333	0.0625	0.357143	0.031746	
3	0.02	Phone	0	0.096774	UPI	Male	0.75	0.6	Mobile	0.5	Single	0.6	1	0.266667	0	0	0	0.372308	0.404682	1	0.09375	0	0.007937	
4	0.02	Phone	0	0.806452	'Debit Carc	Male	0.5	0.6	Mobile	0.5	Single	0.5	1	0.2	0	0	0.166667	0.369231	0.401338	0.933333	0.0625	0.857143	0.007937	
5	0	Phone	1	0.322581	'Debit Carc	Male	0.5	0.6	'Laptop & A	1	Single	0.7	0	0.8	0	0	0.166667	0.412308	0.448161	0.300353	0.104167	0.288895	0	
6	0	Phone	0	0.225806	CC	Male	0	0.4	Mobile	1	Single	0.2	0	0	0.2	0	0.166667	0.4	0.434783	0.300353	0	0.288895	0	
7	0	Computer	0	0.548387	'Debit Carc	Female	0.75	0.8	'Mobile Ph	1	Single	0.1	1	0.733333	0.8	0.625	0.388889	0.427692	0.07748	0.300353	0.9375	0.288895	0	
8	0.02	Phone	1	0.193548	'Cash on D	Male	0.5	0.4	'Laptop & A	0.25	Divorced	0.3	0	0.2	0	0	0	0.372308	0.404682	0.933333	0.041667	0	0.007937	
9	0.26	Phone	1	0.129032	'E wallet'	Male	0	0.6	Mobile	0.5	Divorced	0.1	1	0.2	0	0	0.111111	0.390769	0.424749	0.071795	0	0.043956	0.103175	
10	0.02	Phone	0	0.83871	'Debit Carc	Male	0.5	0.8	Mobile	0.5	Single	0.1	0	0.066667	0.2	0	0.055556	0.378462	0.411371	0.8	0.0625	0.285714	0.007937	
11	0.22	'Mobile Ph	0	0.032258	'Debit Carc	Male	0.75	0.6	Fashion	0.5	Single	0.9	1	0.133333	0	0	0	0.473846	0.51505	0.078788	0.09375	0	0.087302	
12	0	Phone	0	0.193548	COD	Male	0.5	0.4	Mobile	0.5	Single	0.1	1	0.133333	0.4	0.125	0.111111	0.412308	0.22408	0.300353	0.125	0.288895	0	
13	0	Phone	0	0.322581	CC	Male	0.75	0.6	Mobile	0.5	Divorced	0	1	0.4	0	0	0	0.412308	0.448161	0.300353	0.09375	0.288895	0	
14	0.18	'Mobile Ph	1	0.322581	'Credit Car	Male	0.75	0.6	Fashion	0.25	Single	0.1	0	0.333333	0	0.375	0.388889	0.603077	0.16388	0.118519	0.25	0.222222	0.285714	
15	0	Computer	0	0.225806	'Debit Carc	Female	0	0.6	Mobile	0.25	Single	0.1	1	0.466667	0.2	0	0	0.396923	0.431438	0.300353	0	0.288895	0	
16	0	'Mobile Ph	1	0.193548	'E wallet'	Male	0.5	0.6	'Laptop & A	0.5	Single	0.1	1	0	0.2	0	0.166667	0.483077	0.525084	0.300353	0.0625	0.288895	0	
17	0	Computer	0	0.258065	'Debit Carc	Male	0.75	0.8	'Laptop & A	0.5	Single	0.1	1	0.866667	0.2	0	0.333333	0.495385	0.538462	0.300353	0.09375	0.288895	0	
18	0.38	'Mobile Ph	0	0.483871	'Debit Carc	Female	0.75	0.4	'Mobile Ph	0.75	Divorced	0.9	1	0.466667	0.2	0.375	0.166667	0.461538	0.125418	0.063158	0.5	0.045113	0.603175	
19	0	Phone	1	0.677419	'Debit Carc	Female	0.75	0.8	'Laptop & A	0.5	Divorced	0.3	1	0	0.2	0.125	0.111111	0.449231	0.244147	0.300353	0.1875	0.288895	0	
20	0.28	Computer	0	0.290323	'Debit Carc	Male	0.5	0.8	Fashion	0.25	Single	0.2	0	0.266667	0	0	0.166667	0.636923	0.692308	0.071429	0.041667	0.061225	0.111111	
21	0	Phone	0	0.322581	'Credit Car	Female	0.5	0.4	'Mobile Ph	1	Single	0.7	0	0.733333	0.2	0	0.222222	0.436923	0.474916	0.300353	0.104167	0.288895	0	
22	0.16	'Mobile Ph	1	0.032258	'E wallet'	Male	0.75	0.4	Fashion	0.75	Divorced	0.1	0	0.133333	0.2	0	0.333333	0.532308	0.578595	0.108333	0.125	0.214286	0.063492	
23	0.36	Phone	0	0.343222	'Debit Carc	Male	0.5	0.4	'Laptop & A	0.75	Married	0.8	0	0.466667	0.2	0	0.833333	0.378462	0.411371	0.066667	0.083333	0.238095	0.142857	
24	0.1	Computer	1	0.290323	'E wallet'	Female	0.5	0.4	Fashion	0.25	Single	0.1	0	0.2	0.4	0.25	0.388889	0.584615	0.211817	0.186667	0.125	0.4	0.119048	
25	0	'Mobile Ph	0	0.258065	'Credit Car	Male	0.5	0.6	'Laptop & A	0.5	Divorced	0.2	0	0.4	0.2	0	0	0.495385	0.538462	0.300353	0.0625	0.288895	0	
26	0.6	Phone	0	0.322581	CC	Female	0.75	0.6	Mobile	0.75	Divorced	0.4	1	0.6	0.2	0	0	0.409231	0.444816	0.044444	0.125	0	0.238095	
27	0.26	'Mobile Ph	1	0.16129	'E wallet'	Male	0.75	0.6	Fashion	0.25	Divorced	0.7	0	0	0.4	0.125	0.611111	0.643077	0.349498	0.05641	0.125	0.241758	0.206349	
28	0	Phone	0	0.343222	CC	Male	0.75	0.4	Mobile	0.25	Divorced	0.1	0	0.866667	0	0.125	0.111111	0.378462	0.205686	0.300353	0.125	0.288895	0	
29	0.02	Phone	0	0.709677	CC	Male	0	0.4	Mobile	0.25	Divorced	0.1	0	0	0	0	0.111111	0.390769	0.424749	0.733333	0	0.571429	0.007937	
30	0.6	'Mobile Ph	0	0.806452	'Debit Carc	Male	0.5	0.2	Fashion	0.25	Married	0.3	1	0.333333	0.2	0	0.222222	0.596923	0.648829	0.035556	0.041667	0.038095	0.238095	

A4. Data preparation- reduction

After evaluating the full dataset using the **InfoGainAttributeEval** evaluator and **Ranker** method in WEKA. We ranked attributes based on their contribution to predicting the **ChurnCategory** class.

The top-ranked attributes with the highest information gain were:

1. **SatisfactionScore (0.66421)**
2. **EngagementScore (0.28743)**
3. **Tenure (0.20505)**
4. **LoyaltyScore (0.18117)**
5. **OrderHikeRate (0.10324)**

To reduce dimensionality, we manually selected the top 5 attributes as our reduced dataset and removed all attributes that were not part of the 5 by clicking the **Remove** button. This will only leave us with only the top attributes and the class label, aiming to maintain model performance while simplifying the feature set.

Attribute selection output

```
Search Method:
        Attribute ranking.

Attribute Evaluator (supervised, Class (nominal): 24 ChurnCategory):
        Information Gain Ranking Filter

Ranked attributes:
 0.66421   10 SatisfactionScore
 0.28743   21 EngagementScore
 0.20505    1 Tenure
 0.18117   23 LoyaltyScore
 0.10324   20 OrderHikeRate
 0.066     22 InactivityRatio
 0.04674   19 AvgCashbackPerOrder
 0.04318   18 CashbackAmount
 0.03975   13 Complain
 0.03819    9 PreferedOrderCat
 0.03577   17 DaySinceLastOrder
 0.02139   11 MaritalStatus
 0.0191     4 WarehouseToHome
 0.01204    5 PreferredPaymentMode
 0.01117   12 NumberOfAddress
 0.01115    2 PreferredLoginDevice
 0.00781    3 CityTier
 0.00739   14 OrderAmountHikeFromlastYear
 0.0041     7 HourSpendOnApp
 0.00373   15 CouponUsed
 0.00359    8 NumberOfDeviceRegistered
 0.00151    6 Gender
 0         16 OrderCount

Selected attributes: 10,21,1,23,20,22,19,18,13,9,17,11,4,5,12,2,3,14,7,15,8,6,16 : 23
```

The original dataset contains **4447 instances.** To test model generalizability on a smaller dataset, we applied a **Resample** filter with **50%** on sample size. This resulted in a dataset with approximately **2223 instances,** which we used to create a reduced version of the dataset containing only the top attributes and a subset of the data.

Viewer

Relation: A2 (replacemissingvalues).arff-weka.filters.unsupervised.attribute.ReplaceMissingValues-weka.filters.unsupervis

No.	1: Tenure Numeric	2: SatisfactionScore Numeric	3: OrderHikeRate Nominal	4: EngagementScore Numeric	5: LoyaltyScore Numeric	6: **ChurnCategory** Nominal
1	0.1	1.0	Medium	0.833333	0.15873	Not Churn
2	0.26	0.5	Medium	0.0625	0.103175	Not Churn
3	0.3	0.25	Medium	0.166667	0.238095	Uncertain
4	0.18	0.5	Medium	0.09375	0.071429	Not Churn
5	0.28	0.75	Low	0.375	0.333333	Not Churn
6	0.56	0.75	Low	0.25	0.444444	Not Churn
7	0.36	1.0	Low	0.104167	0.142857	Not Churn
8	0.04	0.25	High	0.041667	0.031746	Uncertain
9	0.0	0.5	High	0.09375	0.0	Churn
10	0.6	1.0	Low	0.625	0.714286	Not Churn
11	0.0	0.75	High	0.083333	0.0	Churn
12	0.16	0.5	Medium	0.125	0.126984	Not Churn
13	0.02	0.5	High	0.1875	0.015873	Not Churn
14	0.36	0.5	Medium	0.46875	0.714286	Not Churn
15	0.02	0.5	High	0.375	0.02381	Churn
16	0.02	0.75	High	0.25	0.015873	Churn
17	0.18	0.75	Medium	0.125	0.071429	Not Churn
18	0.16	0.25	Medium	0.125	0.126984	Uncertain
19	0.06	1.0	High	0.3125	0.047619	Not Churn
20	0.06	1.0	High	0.15625	0.02381	Not Churn

Result Comparison

Model	Dataset	Accuracy	Precision	Recall	F1-Score	ROC Area
Trees (J48)	Full Dataset [percentage split **(80:30)**]	0.936 (93.58%)	0.936	0.936	0.936	0.972
	Reduced Dataset [percentage split **(80:30)**]	0.890 (88.98%)	0.890	0.890	0.890	0.945
Naïve Bayes	Full Dataset [20 k-fold]	0.783 (78.30%)	0.832	0.783	0.799	0.908
	Reduced Dataset [percentage split **(80:30)**]	0.822 (82.24%)	0.869	0.822	0.838	0.929

B.Model Development and Evaluation

Model: Trees (J48)

Three different evaluation approaches were conducted:

1. 10-Fold Cross-Validation

The dataset was divided into 10 subsets. The model was trained on 9 and tested on 1, repeating the process 10 times.

Correctly Classified Instances	4152	**93.3663 %**
Incorrectly Classified Instances	295	**6.6337 %**
Root mean squared error		**0.1949**
Relative absolute error		**12.924 %**
Root relative squared error		**43.9369 %**

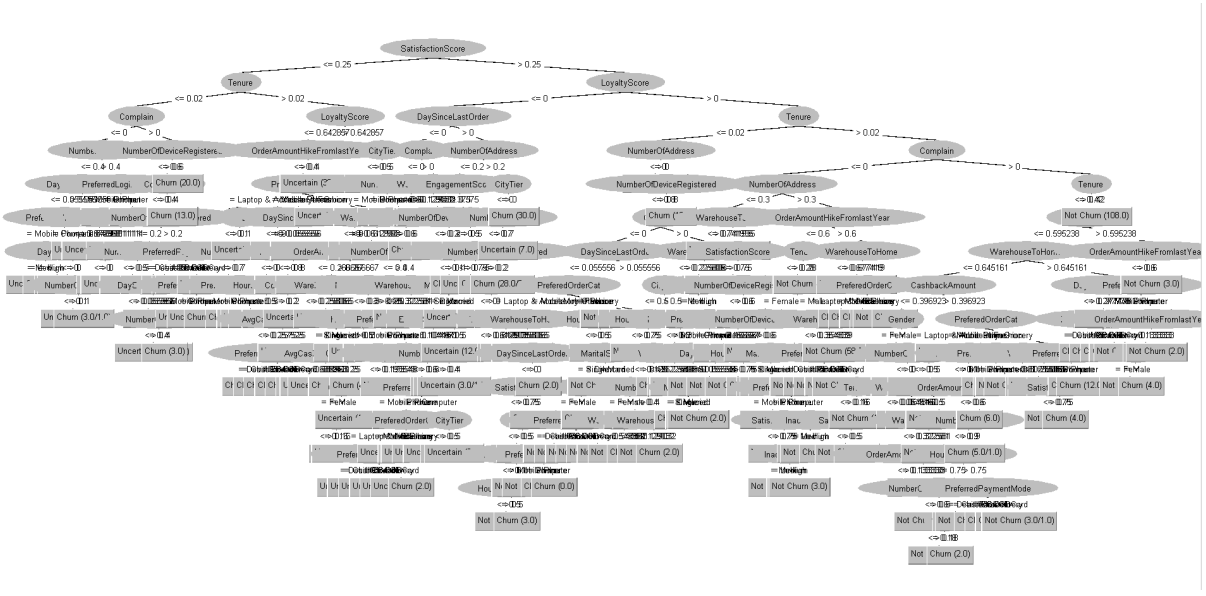
Class	Accuracy	Precision	Recall	F1-Score	ROC Area
Not Churn	0.973	0.963	0.973	0.968	0.987

Uncertain	0.957	0.944	0.957	0.951	0.983
Churn	0.736	0.789	0.736	0.761	0.930
Weighted Avg.	0.934	0.932	0.934	0.933	0.977

Confusion Matrix

a	**b**	**c**	
2311	0	65	**a = Not Churn**
0	1371	61	**b = Uncertain**
88	81	470	**c = Churn**

Tree Visualization:



2. 20-Fold Cross-Validation

Similar to the previous method, but using 20 folds.

Correctly Classified Instances	4158	**93.5012 %**
Incorrectly Classified Instances	289	**6.4988 %**
Root mean squared error		**0.1909**

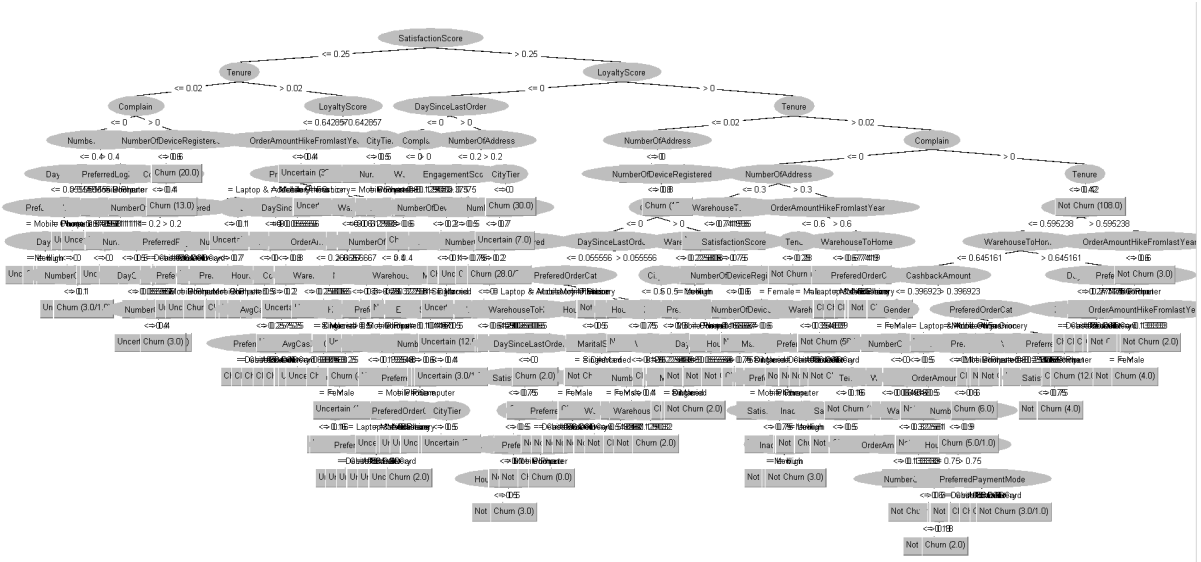
Relative absolute error		**12.0042 %**
Root relative squared error		**43.0449 %**

Class	Accuracy	Precision	Recall	F1-Score	ROC Area
Not Churn	0.970	0.960	0.970	0.965	0.985
Uncertain	0.957	0.957	0.957	0.957	0.990
Churn	0.756	0.784	0.756	0.770	0.949
Weighted Avg.	0.935	0.934	0.935	0.935	0.982

Confusion Matrix

a	b	c	
2304	0	72	**a = Not Churn**
0	1371	31	**b = Uncertain**
95	61	483	**c = Churn**

Tree Visualization:



3. Percentage Split

Two splits were tested: 70% training / 30% testing, and 80% training / 20% testing using the "Percentage split" option in WEKA.

i. 70:30 (70% training / 30% testing)

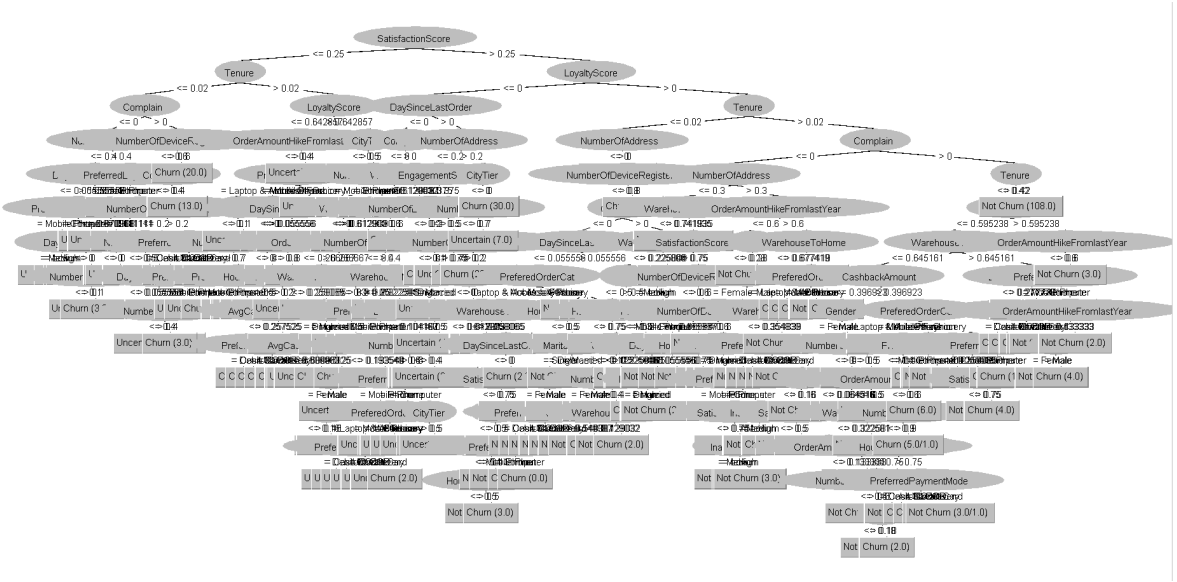
Correctly Classified Instances	1229	**92.1289 %**
Incorrectly Classified Instances	105	**7.8711 %**
Root mean squared error		**0.2167**
Relative absolute error		**17.8063 %**
Root relative squared error		**49.0139 %**

Class	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**ROC Area**
Not Churn	0.958	0.964	0.958	0.961	0.975
Uncertain	0.978	0.910	0.978	0.942	0.981
Churn	0.673	0.770	0.673	0.718	0.843
Weighted Avg.	0.921	0.919	0.921	0.919	0.957

Confusion Matrix

a	b	c	
703	0	31	**a = Not Churn**
0	392	9	**b = Uncertain**
26	39	134	**c = Churn**

Tree Visualization:



ii. 80:30 (80% training / 20% testing)

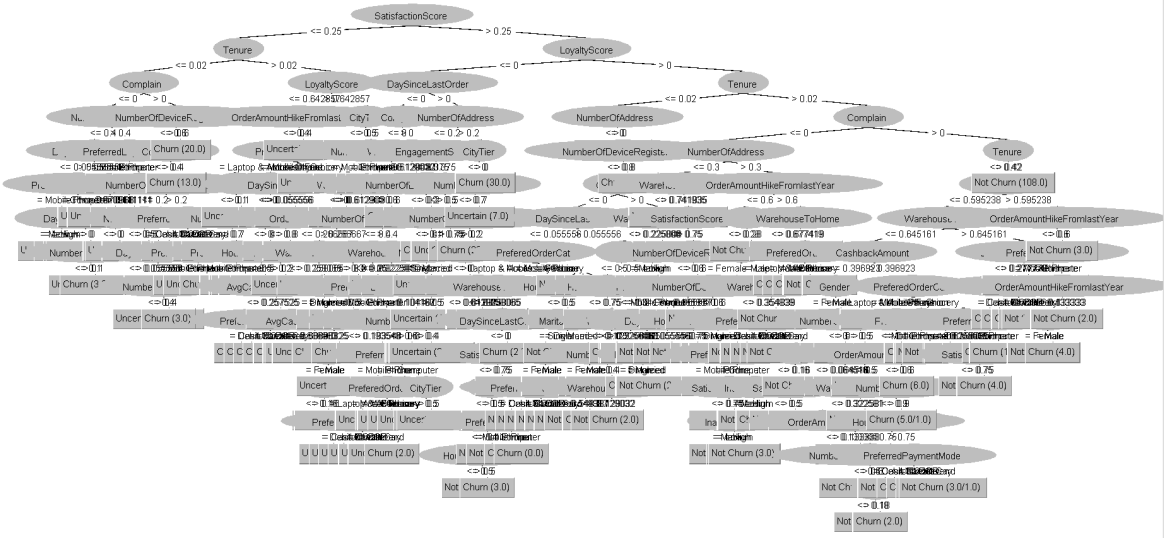
Correctly Classified Instances	832	**93.5883 %**
Incorrectly Classified Instances	57	**6.4117 %**
Root mean squared error		**0.1947**
Relative absolute error		**13.4616 %**
Root relative squared error		**44.0106 %**

Class	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**ROC Area**
Not Churn	0.959	0.975	0.959	0.967	0.982
Uncertain	0.973	0.934	0.973	0.953	0.984
Churn	0.779	0.797	0.779	0.788	0.914
Weighted Avg.	0.936	0.936	0.936	0.936	0.972

Confusion Matrix

a	**b**	**c**	
470	0	20	**a = Not Churn**
0	256	7	**b = Uncertain**
12	18	106	**c = Churn**

Tree Visualization:


SatisfactionScore
<= 0.25
> 0.25
Tenure
LoyaltyScore
<= 0.02
> 0.02
<= 0
> 0
Complain
LoyaltyScore
DaySinceLastOrder
Tenure
<= 0
> 0
<= 0.64285
<= 0
> 0
<= 0.02
> 0.02
NumberOfAddress
NumberOfAddress
Complain
<= 0.4
<= 0
> 0
Churn (20.0)
Churn (13.0)
Churn (30.0)
Not Churn (108.0)
<= 0.2
> 0.2
Uncertain (7.0)
<= 0.055556
> 0.055556
<= 0.741935
> 0.741935
<= 0.6
> 0.6
<= 0.595238
> 0.595238
<= 0.645161
> 0.645161
Not Churn (3.0)
Churn (3.0)
Churn (3.0)
Churn (2.0)
Not Churn (2.0)
Not Churn (4.0)
Churn (6.0)
Not Churn (4.0)
Churn (2.0)
Churn (5.0/1.0)
Churn (2.0)
Not Churn (0.0)
Not Churn (3.0)
Not Churn (3.0/1.0)
Churn (2.0)

Model: Naïve Bayes

Three different evaluation approaches were conducted:

1. 10-Fold Cross-Validation

The dataset was divided into 10 subsets. The model was trained on 9 and tested on 1, repeating the process 10 times.

Correctly Classified Instances	3475	**78.1426 %**
Incorrectly Classified Instances	972	**21.8574 %**
Root mean squared error		**0.3412**
Relative absolute error		**44.2947 %**
Root relative squared error		**76.9156 %**

Class	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**ROC Area**
Not Churn	0.799	0.915	0.799	0.853	0.920
Uncertain	0.786	0.877	0.786	0.829	0.929
Churn	0.706	0.414	0.706	0.522	0.818
Weighted Avg.	0.781	0.831	0.781	0.798	0.908

Confusion Matrix

a	**b**	**c**	
1898	91	387	**a = Not Churn**
55	1126	251	**b = Uncertain**
121	67	451	**c = Churn**

2. 20-Fold Cross-Validation

Similar to the previous method, but using 20 folds.

Correctly Classified Instances	3480	**78.255 %**
Incorrectly Classified Instances	967	**21.745 %**
Root mean squared error		**0.3412**
Relative absolute error		**44.3068 %**
Root relative squared error		**76.9229 %**

Class	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**ROC Area**
Not Churn	0.799	0.917	0.799	0.854	0.920
Uncertain	0.788	0.878	0.788	0.831	0.929
Churn	0.707	0.415	0.707	0.523	0.818
Weighted Avg.	0.783	0.832	0.783	0.799	0.908

Confusion Matrix

a	**b**	**c**	
1899	90	387	**a = Not Churn**
53	1129	250	**b = Uncertain**
120	67	452	**c = Churn**

3. Percentage Split

Two splits were tested: 70% training / 30% testing, and 80% training / 20% testing using the "Percentage split" option in WEKA.

i. 70:30 (70% training / 30% testing)

Correctly Classified Instances	1029	**77.1364 %**
Incorrectly Classified Instances	305	**22.8636 %**
Root mean squared error		**0.3443**
Relative absolute error		**44.9082 %**
Root relative squared error		**77.89 %**

Class	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**ROC Area**
Not Churn	0.767	0.080	0.767	0.837	0.920
Uncertain	0.808	0.069	0.808	0.821	0.935
Churn	0.714	0.424	0.714	0.532	0.827
Weighted Avg.	0.771	0.821	0.771	0.787	0.911

Confusion Matrix

a	**b**	**c**	
563	38	133	**a = Not Churn**
17	324	60	**b = Uncertain**
31	26	142	**c = Churn**

ii. 80:30 (80% training / 20% testing)

Correctly Classified Instances	680	**76.4904 %**
Incorrectly Classified Instances	209	**23.5096 %**
Root mean squared error		**0.3555**
Relative absolute error		**46.7117 %**

Root relative squared error		**80.3669 %**

Class	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**ROC Area**
Not Churn	0.776	0.909	0.776	0.837	0.909
Uncertain	0.791	0.835	0.791	0.813	0.924
Churn	0.676	0.414	0.676	0.514	0.797
Weighted Avg.	0.765	0.812	0.765	0.780	0.896

Confusion Matrix

a	**b**	**c**	
380	22	88	**a = Not Churn**
13	208	42	**b = Uncertain**
25	19	92	**c = Churn**

After data reduced:

Model: Trees (J48)

Three different evaluation approaches were conducted:

1. 10-Fold Cross-Validation

The dataset was divided into 10 subsets. The model was trained on 9 and tested on 1, repeating the process 10 times.

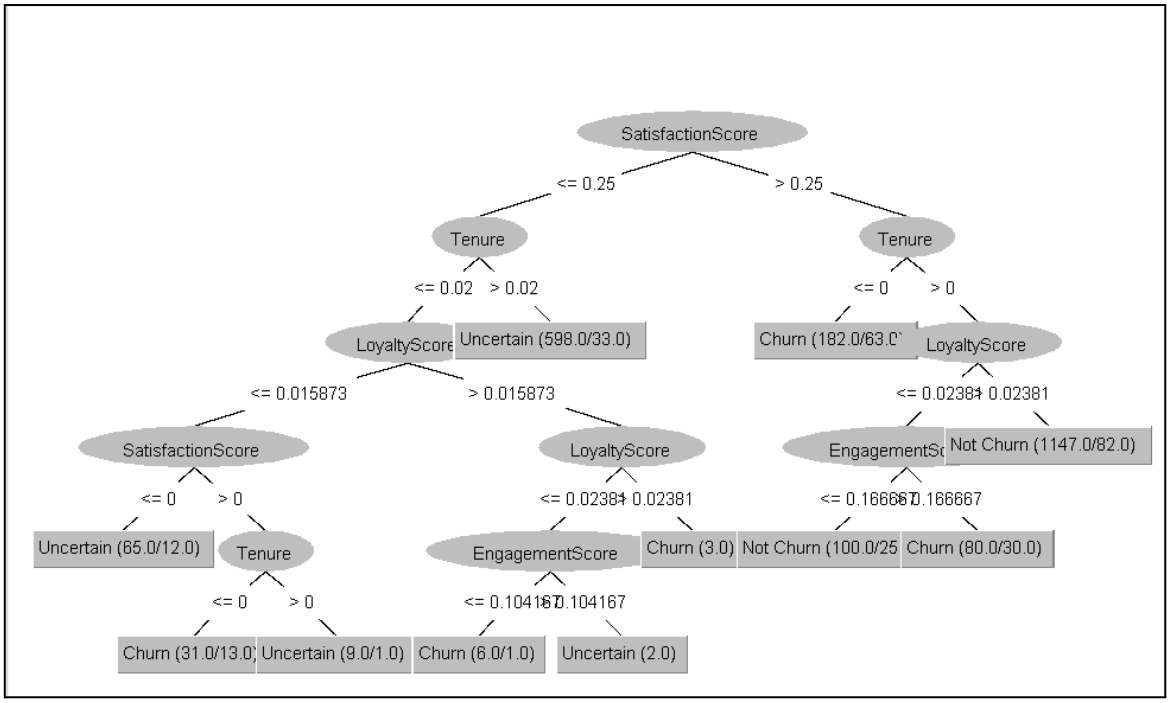
Correctly Classified Instances	1956	**87.9892 %**
Incorrectly Classified Instances	267	**12.0108 %**
Root mean squared error		**0.1273**
Relative absolute error		**32.0118 %**
Root relative squared error		**57.2961 %**

Class	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**ROC Area**
Not Churn	0.967	0.913	0.967	0.939	0.947
Uncertain	0.904	0.924	0.904	0.914	0.97
Churn	0.537	0.638	0.537	0.583	0.777
Weighted Avg.	0.880	0.873	0.880	0.875	0.929

Confusion Matrix

a	**b**	**c**	
1140	0	39	**a = Not Churn**
0	629	67	**b = Uncertain**
10	52	187	**c = Churn**

Tree Visualization:



2. 20-Fold Cross-Validation

Similar to the previous method, but using 20 folds.

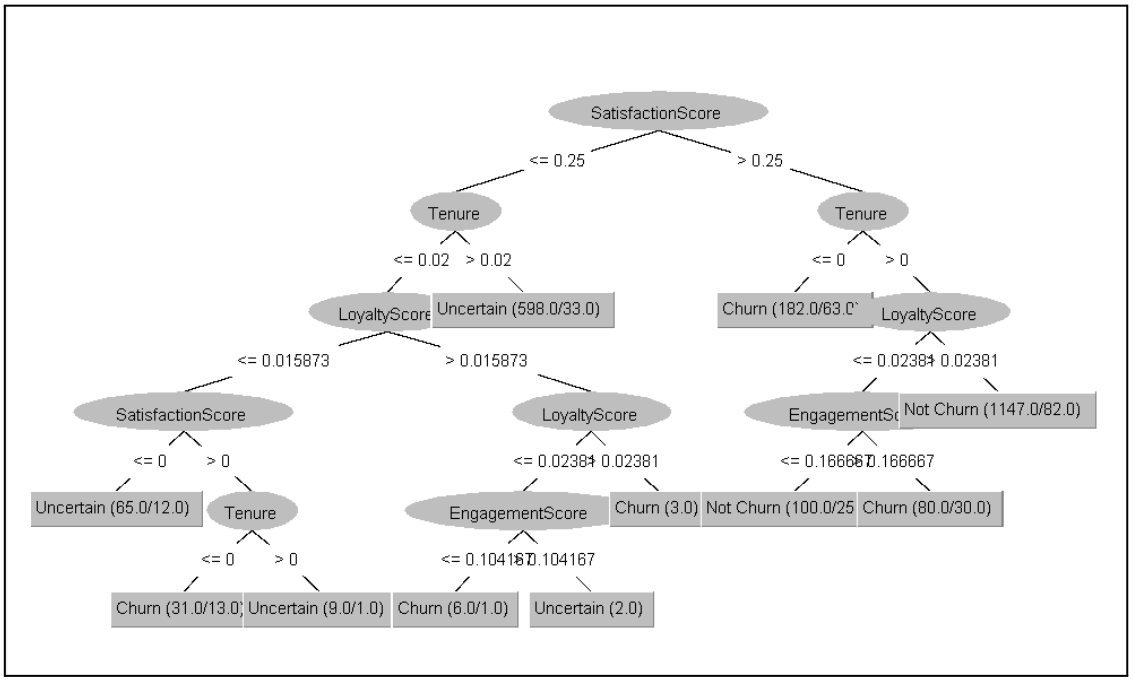
Correctly Classified Instances	1954	**87.8992 %**
Incorrectly Classified Instances	269	**12.1008 %**
Root mean squared error		**0.2552**
Relative absolute error		**32.1128 %**
Root relative squared error		**57.2365 %**

Class	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**ROC Area**
Not Churn	0.964	0.913	0.964	0.938	0.947
Uncertain	0.901	0.928	0.901	0.914	0.975
Churn	0.546	0.631	0.546	0.586	0.774
Weighted Avg.	0.879	0.873	0.879	0.875	0.929

Confusion Matrix

a	b	c	
1137	0	42	**a = Not Churn**
0	627	69	**b = Uncertain**
109	49	190	**c = Churn**

Tree Visualization:



3. Percentage Split

Two splits were tested: 70% training / 30% testing, and 80% training / 20% testing using the "Percentage split" option in WEKA.

i. 70:30 (70% training / 30% testing)

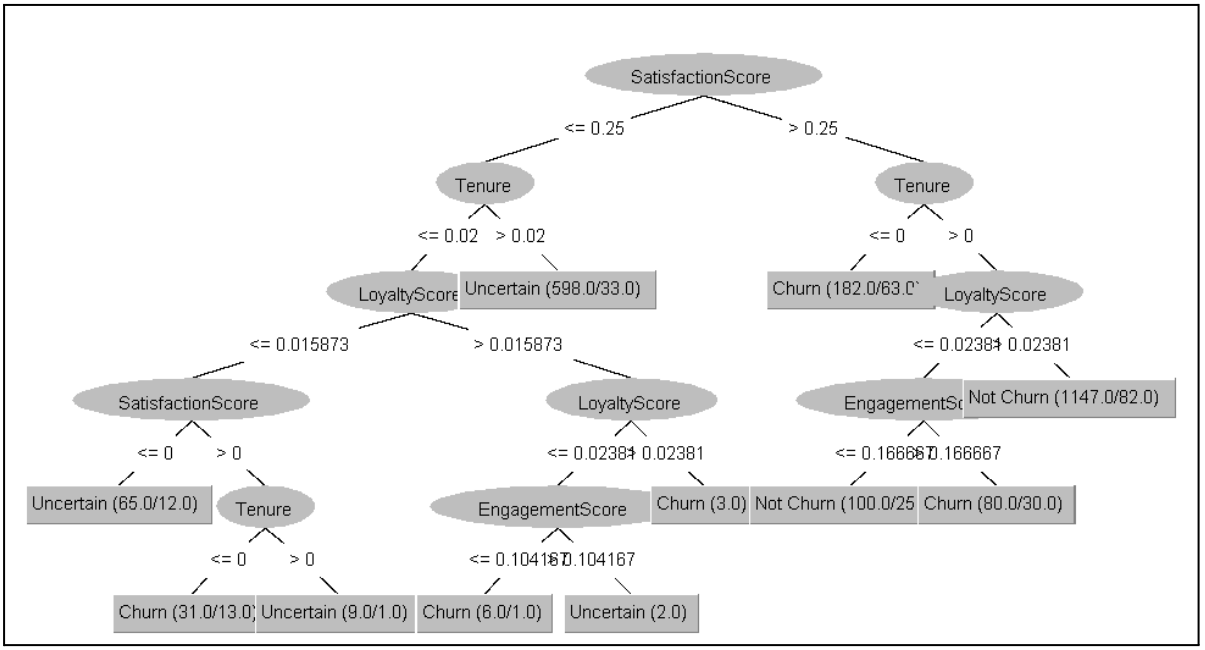
Correctly Classified Instances	580	**86.9565 %**
Incorrectly Classified Instances	87	**13.0435 %**
Root mean squared error		**0.2606**
Relative absolute error		**34.6191 %**
Root relative squared error		**58.8178 %**

Class	Accuracy	Precision	Recall	F1-Score	ROC Area
Not Churn	0.959	0.931	0.959	0.945	0.961
Uncertain	0.896	0.878	0.896	0.887	0.966
Churn	0.500	0.586	0.500	0.540	0.776
Weighted Avg.	0.870	0.862	0.870	0.865	0.934

Confusion Matrix

a	b	c	
349	0	15	**a = Not Churn**
0	180	21	**b = Uncertain**
26	25	51	**c = Churn**

Tree Visualization:



ii. 80:30 (80% training / 20% testing)

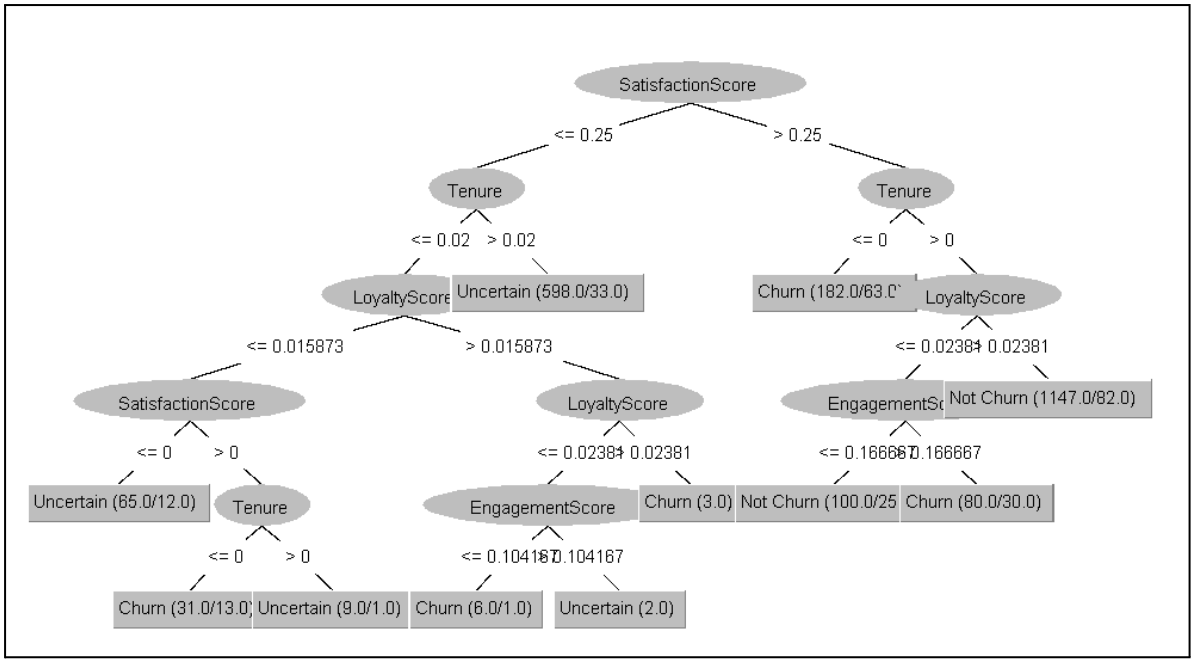
Correctly Classified Instances	396	**88.9888 %**
Incorrectly Classified Instances	49	**11.0112 %**
Root mean squared error		**0.2381**
Relative absolute error		**30.9069 %**
Root relative squared error		**54.0799 %**

Class	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**ROC Area**
Not Churn	0.960	0.940	0.960	0.950	0.964
Uncertain	0.889	0.930	0.889	0.909	0.978
Churn	0.619	0.609	0.619	0.614	0.801
Weighted Avg.	0.890	0.890	0.890	0.890	0.945

Confusion Matrix

a	**b**	**c**	
237	0	10	**a = Not Churn**
0	120	15	**b = Uncertain**
15	9	39	**c = Churn**

Tree Visualization:



Model: Naïve Bayes

Three different evaluation approaches were conducted:

1. 10-Fold Cross-Validation

The dataset was divided into 10 subsets. The model was trained on 9 and tested on 1, repeating the process 10 times.

Correctly Classified Instances	1763	**79.3072 %**
Incorrectly Classified Instances	460	**206928 %**
Root mean squared error		**0.3325**
Relative absolute error		**46.4611 %**
Root relative squared error		**74.5959 %**

Class	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**ROC Area**
Not Churn	0.797	0.919	0.797	0.854	0.933
Uncertain	0.868	0.906	0.868	0.886	0.933
Churn	0.629	0.411	0.629	0.497	0.794
Weighted Avg.	0.793	0.835	0.793	0.808	0.911

Confusion Matrix

a	b	c	
940	0	239	**a = Not Churn**
17	604	75	**b = Uncertain**
66	63	219	**c = Churn**

2. 20-Fold Cross-Validation

Similar to the previous method, but using 20 folds.

Correctly Classified Instances	1765	**79.3972 %**
Incorrectly Classified Instances	458	**20.6028 %**
Root mean squared error		**0.3326**
Relative absolute error		**46.454 %**
Root relative squared error		**74.6017 %**

Class	Accuracy	Precision	Recall	F1-Score	ROC Area
Not Churn	0.798	0.919	0.798	0.854	0.933
Uncertain	0.871	0.904	0.871	0.887	0.933
Churn	0.626	0.412	0.626	0.497	0.793
Weighted Avg.	0.794	0.835	0.794	0.809	0.911

Confusion Matrix

a	b	c	
941	0	238	**a = Not Churn**
17	606	73	**b = Uncertain**
66	64	218	**c = Churn**

3. Percentage Split

Two splits were tested: 70% training / 30% testing, and 80% training / 20% testing using the “Percentage split” option in WEKA.

i. **70:30 (70% training / 30% testing)**

Correctly Classified Instances	530	**79.4603 %**
Incorrectly Classified Instances	137	**20.5397 %**
Root mean squared error		**0.3256**
Relative absolute error		**46.7216 %**
Root relative squared error		**73.4903 %**

Class	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**ROC Area**
Not Churn	0.797	0.942	0.797	0.863	0.949
Uncertain	0.856	0.891	0.856	0.873	0.935
Churn	0.667	0.410	0.667	0.507	0.824
Weighted Avg.	0.795	0.845	0.795	0.812	0.926

Confusion Matrix

a	**b**	**c**	
290	0	74	**a = Not Churn**
5	172	24	**b = Uncertain**
13	21	68	**c = Churn**

ii. **80:30 (80% training / 20% testing)**

Correctly Classified Instances	366	**82.2472 %**
Incorrectly Classified Instances	79	**17.7528 %**
Root mean squared error		**0.3115**
Relative absolute error		**43.997 %**
Root relative squared error		**70.7743 %**

Class	Accuracy	Precision	Recall	F1-Score	ROC Area
Not Churn	0.826	0.949	0.826	0.883	0.949
Uncertain	0.859	0.921	0.859	0.889	0.936
Churn	0.730	0.442	0.730	0.551	0.836
Weighted Avg.	0.822	0.869	0.822	0.838	0.929

Confusion Matrix

a	b	c	
204	0	43	**a = Not Churn**
4	116	15	**b = Uncertain**
7	10	46	**c = Churn**

Result Comparison

	Model	Methods	Accuracy	Precision	Recall	F1-Score	ROC Area
Before reduced	Trees (J48)	10 fold	0.934	0.932	0.934	0.933	0.977
		20 fold	0.935	0.934	0.935	0.935	0.982
		70:30	0.921	0.919	0.921	0.919	0.957
		80:20	0.936	0.936	0.936	0.936	0.972
	Naïve Bayes	10 fold	0.781	0.831	0.781	0.798	0.908
		20 fold	0.783	0.832	0.783	0.799	0.908
		70:30	0.771	0.821	0.771	0.787	0.911
		80:20	0.765	0.812	0.765	0.780	0.896

	Model	Methods	Accuracy	Precision	Recall	F1-Score	ROC Area
After reduced	Trees (J48)	10 fold	0.880	0.873	0.880	0.875	0.929
		20 fold	0.879	0.873	0.879	0.875	0.929
		70:30	0.870	0.862	0.870	0.865	0.934
		80:20	0.890	0.890	0.890	0.890	0.945
	Naïve Bayes	10 fold	0.793	0.835	0.793	0.808	0.911
		20 fold	0.794	0.835	0.794	0.809	0.911
		70:30	0.795	0.845	0.795	0.812	0.926
		80:20	0.822	0.869	0.822	0.838	0.929